Technique used for Garment dyeing

There are various technique used for garment dyeing; recently some latest technology has developed for garment dyeing in our industry.

1. Tie dyeing
2. Dip dyeing
3. Spray dyeing
4. Over dyeing
5. Reverse dyeing
6. Top dyeing

Tie dyeing

Tie-dye is a process of dyeing textiles or clothing which is made from knit or woven fabric, usually cotton; typically using bright colors. Tie dye is a popular technique used to create colorful designs and patterns on clothing. Various techniques have been used to tie dye clothing for more than 6000 years. Today, tie dye is a great way to jazz up plain t-shirts and other garments using colorfast dyes and creative folding techniques. Tie-dying became fashionable in the late 1960s & early 1970s as part of hippie style. It was popularized in the U.S. by musicians such as John Sebastian and Janis Joplin. It is same as that of batik printing but here the dye is resisted by knots that are tied in the cloth before it is immersed in dye bath. The outside of the knotted portion is dyed, but inside is not penetrated if the knot is firmly tied. This gives a characteristic blurred or mottled effect.

Advantage of Tie dyeing

- Interesting design created on fabric
- No m/c cost is there

Disadvantage of Tie dyeing

- Costly
- Laborious
- Time taking
- Skilled labor required

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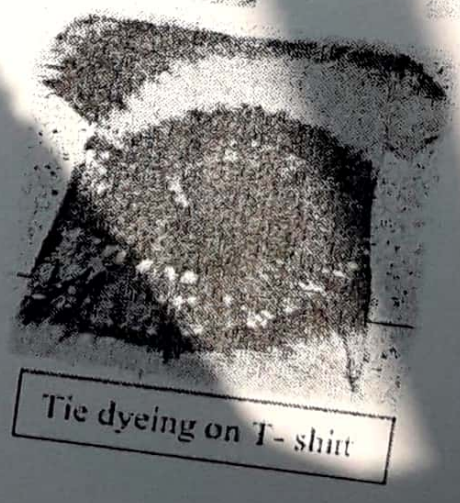
Hand tie dye cotton tunic



Hand crush tie dye tunic kaftan with hand bead work





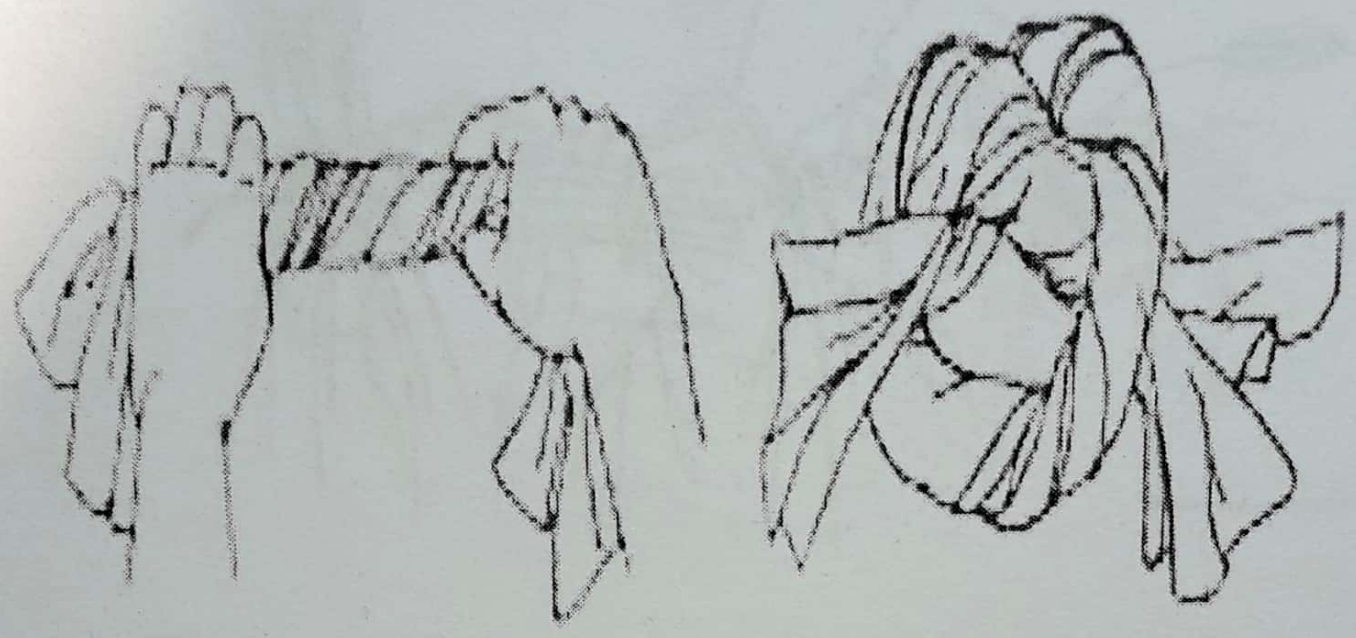


Tie dyeing on T- shirt

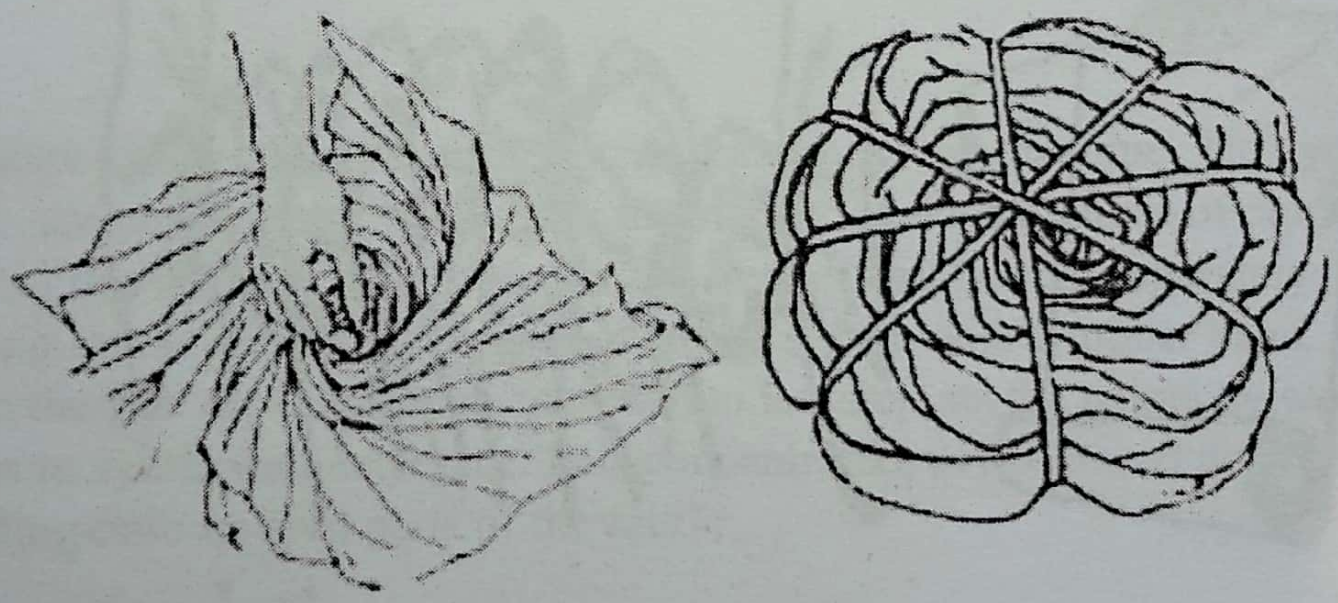
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Tying techniques

Knot Tying

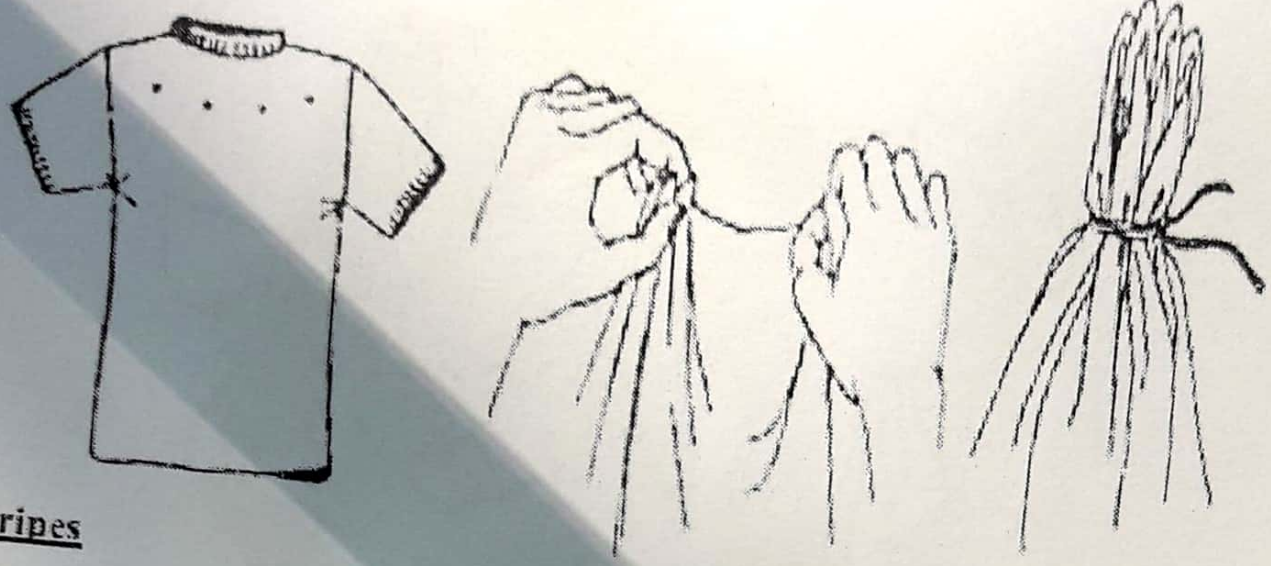


Spirals

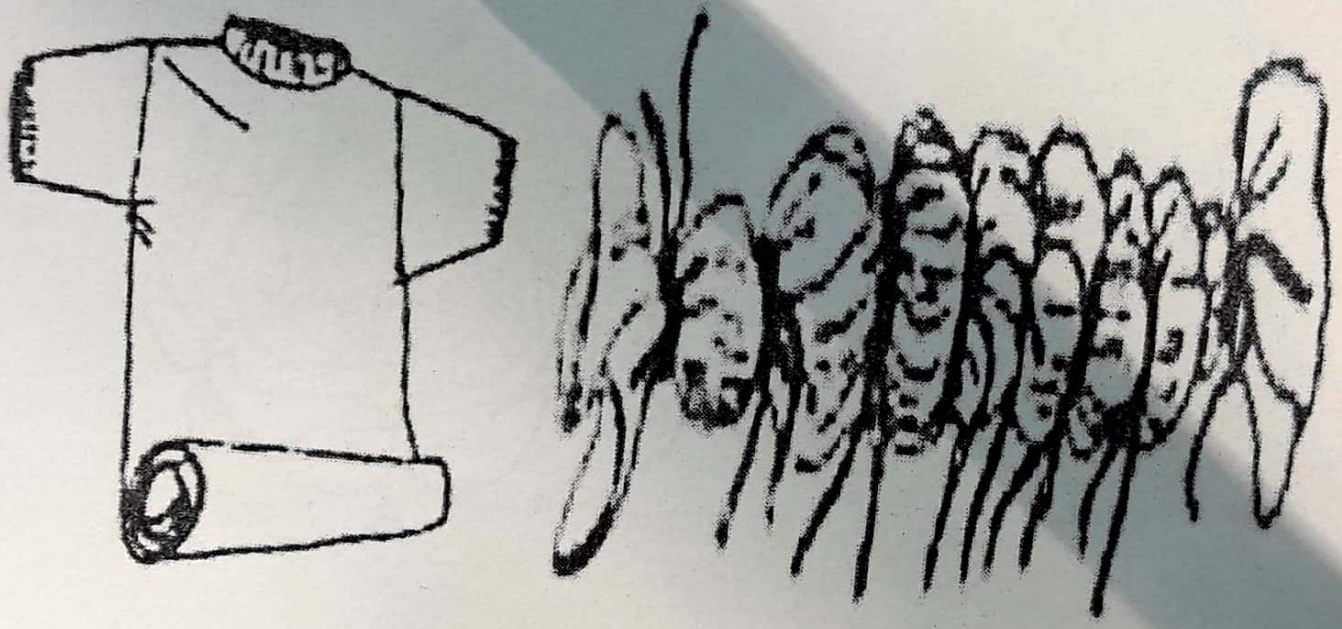


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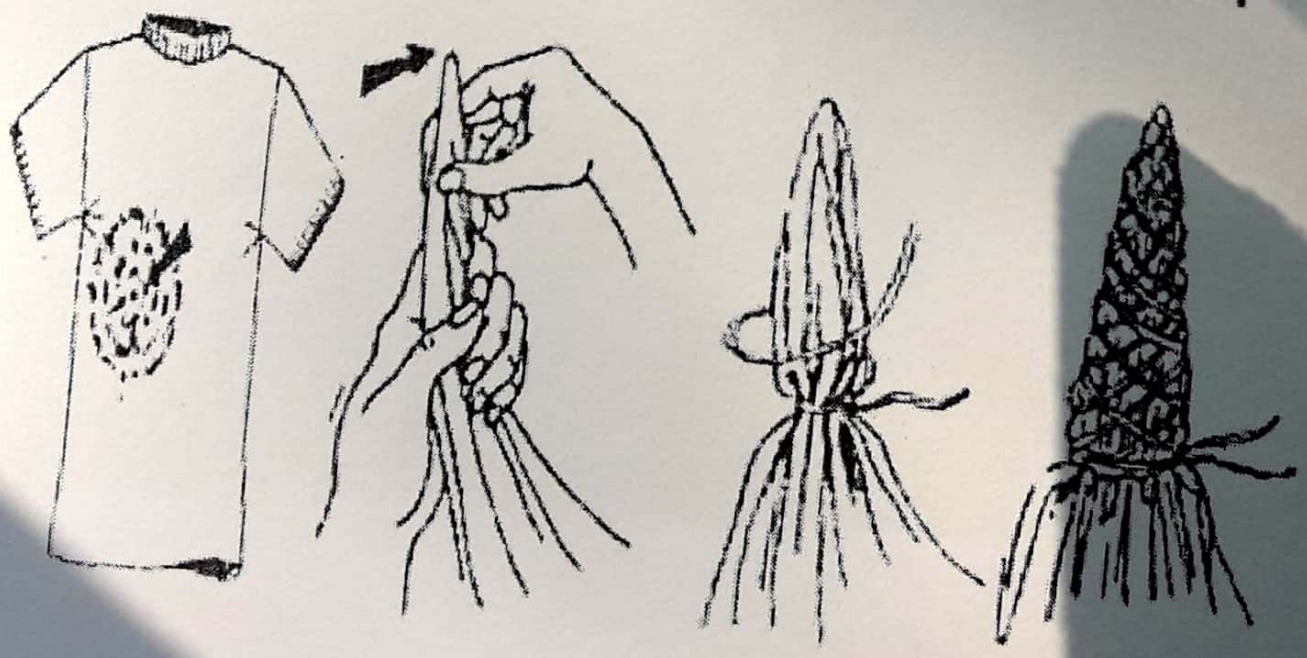
Rosettes



Stripes

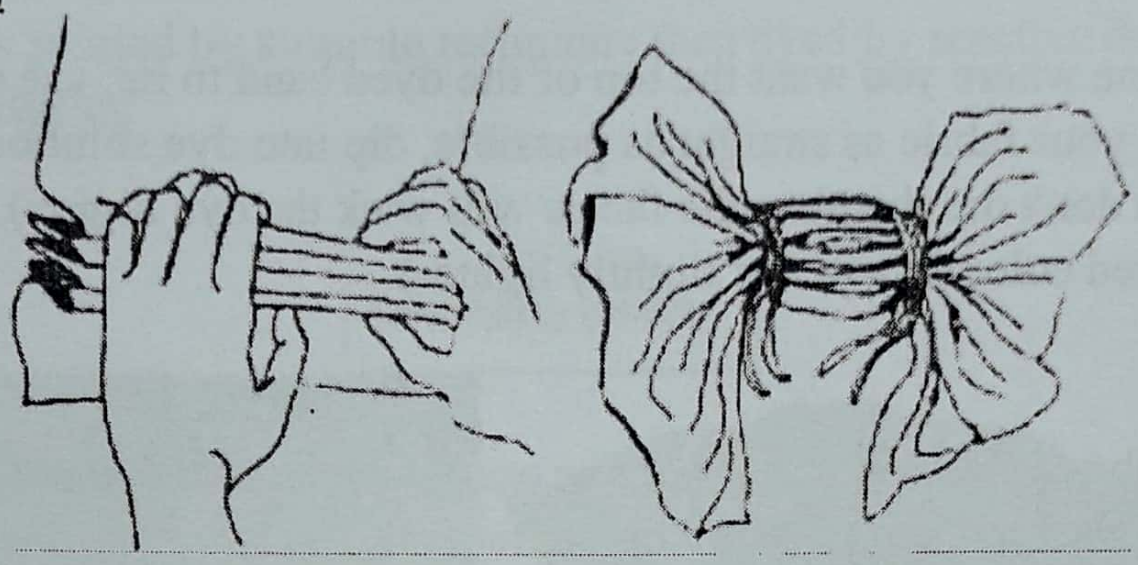


Circles



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Pleats



Dip dyeing

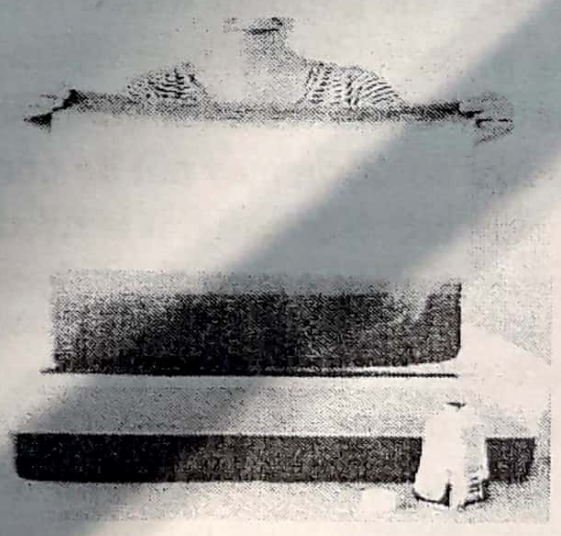
If somebody wants to dip dye of a garments need a dip dyeing machine. Tie dyeing makes every garment unique, so garment specialist gives the steps to make any piece of clothing special with the proper way of dip dyeing. It's best to dye the sleeves and bottom separately: Roll up the part you're not dyeing into a plastic bag to protect it, and secure with a rubber band. Dye, rinse, and let dry between the two dips. For items you plan to launder, set the colors by washing the finished objects in synthrapol detergent. This concentrated, pH-balanced detergent will keep the dye from bleeding onto undyed parts of the fabric.

Dip dyeing technique

Steps of dip dyeing

Step 1

Wash and dry the fabric or garment to remove any finishes that resist dye. Fill plastic bin with hot water to a depth of at least 6 inches. Stir in dye, starting with a small amount and adding more as desired. Dip a sample strip of fabric or paper towel to test the color. If too dark, add water; if too light, add dye. Let solution settle, so there are no bubbles at the surface.



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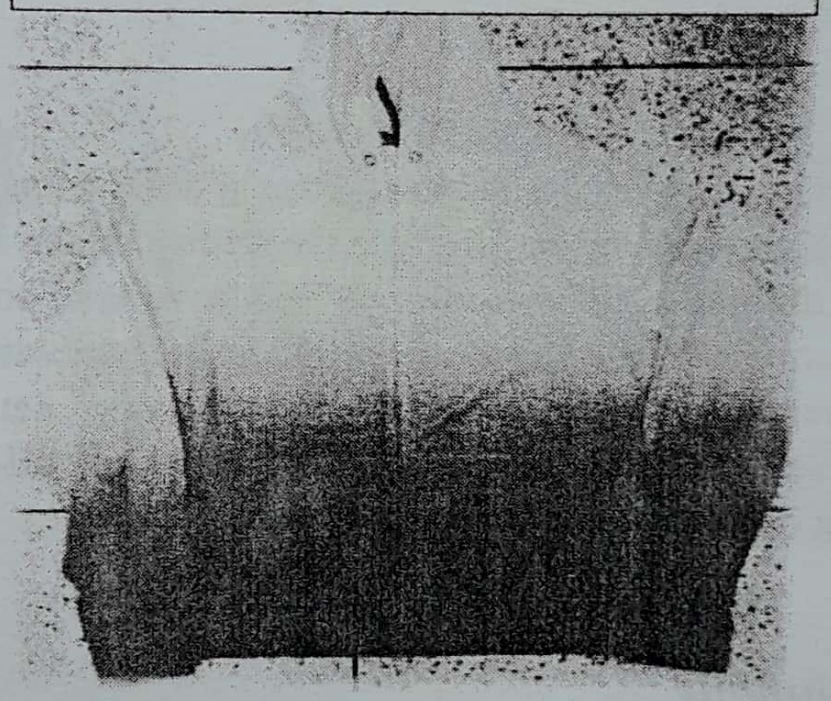
Step 2

Determine where you want the top of the dyed band to be; use pins to mark that line on fabric. Holding your fabric as straight as possible, dip into dye solution to just below the pin line (make sure you don't dip the pins; the fabric will wick the dye higher). Keep fabric in dye until it turns the desired color (it will dry slightly lighter).

Step 3

Rinse fabric in cold water to remove excess dye until the water runs clear. Hang to dry. If desired, wash item with synthrapol detergent.

Dip dyeing hoody shirt/jacket



Over dyeing

(Over Dyeing) to dye for a second or third time with a different color Over dyeing is such a rewarding way of rescuing an ugly or unsatisfactory colored cloth. It gives uneven look. Over dyeing doesn't mean all-time dyeing the garment which is previously dyed. Mainly over dyeing is normal dyeing or piece dyeing process. But in over dyeing process sometimes printed garment is dyeing, basically pigment printed. This method is the best process of the dyeing of garments. However, the penetration of the dye solution may not be completely passed to the fibers such as between the seams, buttons, zippers etc. Normally, it is used for lingerie, socks, sweater dyeing etc. In over dyeing process reactive, direct and pigment dyes are commonly used.

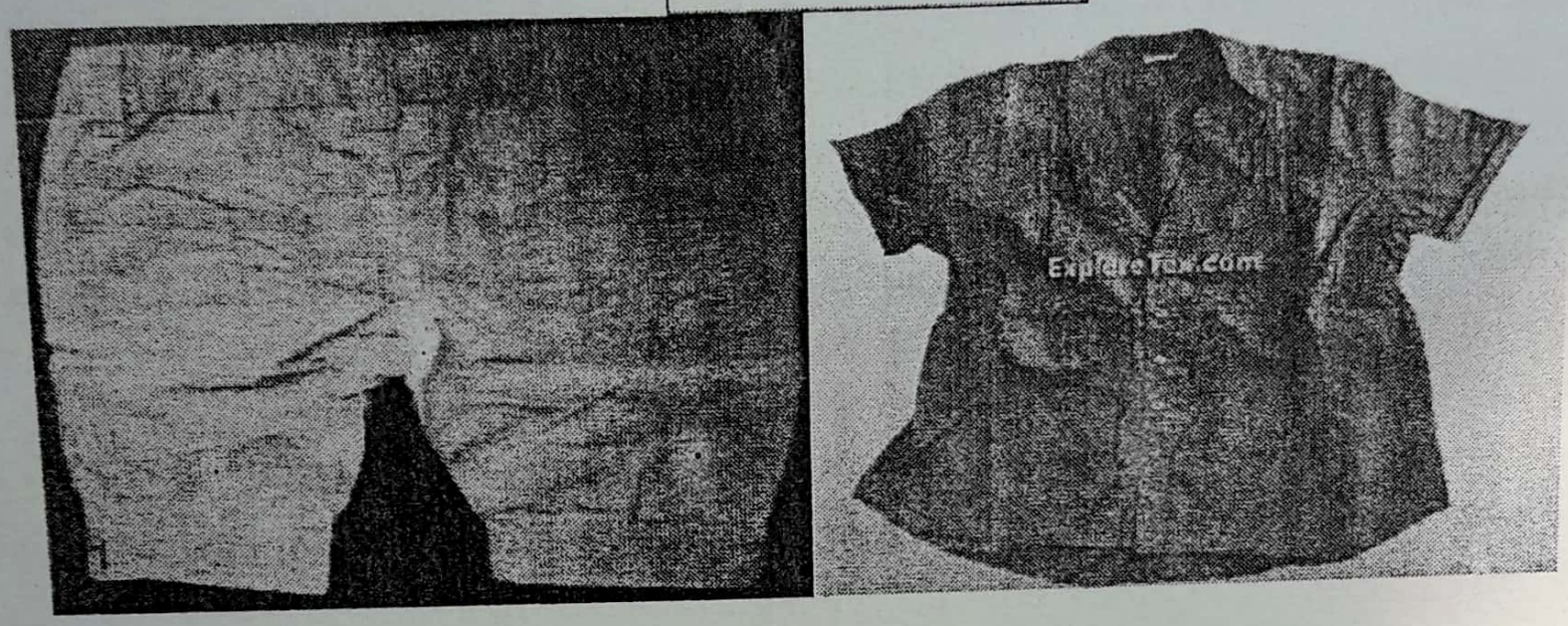
- ❖Two times garments dyeing are called over dyeing.
- ❖ First garment is dyed by pigment dyes then direct dyes this is also called over dyeing.
- ❖ Tinting by any kind of toner is also called over dyeing.
- ❖ Ready for dyeing (RFD) is dyed by any color is called over dyeing.
- ❖ Fabric is dyed in any color then make garment without accessories finally after making

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garment again dyed by same dyes is also called over dyeing

❖ Ready for dyeing(RFD) is printed by suitable technique then dyed by reactive or direct or pigment dyes is called over dyeing.

Over dyeing garment



Problems Related to Garment Dyeing

Though garment dyeing has attracted many processors to look forward, the process is attached with many stringent requirements related to seams, elasticated areas, waist band, cuffs and problems often occur due to shrink behaviour, chafe marks / creases, accessories, sewing threads, foreign substances, interlining and care labeling. Processing problems related to the garment size control and appearance normally result from variations in yarn size / twist, ends per inch, picks per inch or courses per inch in the knitted fabrics. Preparation is an area of textiles in which the least attention is paid to correct procedures. Most of all dyeing problems can be traced to improperly prepared cloth. Natural impurities in the cotton vary among the various locations across the world. Shade non-uniformity and the related appearance problems arise due to the differences in fabric preparation processes like, desizing, scouring and bleaching. Retention of shade, in the case of rayon fabrics, is unstable than that of cotton fabrics. All these problems are further aggravated, if there is any inconsistency in the garment assembly. Though garment dyeing has been reviewed by many authors, the problems arising in the garment dyeing are discussed in detail .Some of the problems that are difficult to correct after dyeing include: poor fabric preparation, improper tension of the sewing threads that may lead to seam puckering, garments prepared with panels taken from differently or inconsistently prepared fabrics and discoloration of button and closures caused by dyes and chemicals used in the process. Consistent shade replication depends on material to liquor ratio, salt concentration, heating rate, alkali concentration, fixation time and temperature. Chlorine emitted from bleaching or ice wash can lead to white specks. Also, the residual hydrogen peroxide in the bleached materials can interfere with dyes and spoil the colours.

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Different machines used in the fabric preparation can result in differences due to the time and temperature differences. While sewing, it is necessary to keep sewing tensions minimum on lock stitch and chain stitch operations. About 90 – 95% of the garments that the dyer receives are cut and sewn from previously prepared cloth. Oils, softeners used in knitting needs to be removed. The various problems observed in the dyed garments include variation in shade / depth within the garment panels, between garments occurs due to mixing of panels cut from differently treated fabrics, white undyed / light seams happens mainly due to tight stitching, which further gets tighter due to the shrinkage during a higher temperature of dyeing, and prevents dye penetration on seams and underneath the stitches, variation in the shade and brightness differences within the garment panels happens due mixing fabrics treated with OBA and without OBA, finish, resin and etc has been added to the garments. Dimensionally stable thread with low elongation will help prevent seam puckering after dyeing. The dimensional stability of the stitching thread should be tested prior to stitching and thread with low elongation can prevent seam puckering after dyeing. Core spun thread also give excellent results. In the case of knitted fabrics, pre relaxation is employed to avoid the problems of seam pucker and garment distortion and such fabrics can also be used along with woven fabrics as fabric cut and sewn garment designs.

Size of the filaments also affects the depth of the shade, due to the fact that coarser filaments absorb more light than finer filaments. As the cross section of fibres becomes more and more non-circular, the fibres light scattering power increases. Trilobal fibres need about 50% extra dye than circular fibres for same depth. However, bulked multi-lobal fibres in the yarn —trap the light within the body of the textile and increases the depth of the colour. Therefore, light scattering or light trapping can alter the depth as decided by cross section.

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